Validation for Sybil

Research Objective(一):

Sybil validation and External validation.

Dataset(一):

TCIA NLST data set

Cohort Selection(一二):

TCIA NLST dataset patients with or without cancer.

DICOM QC(二):

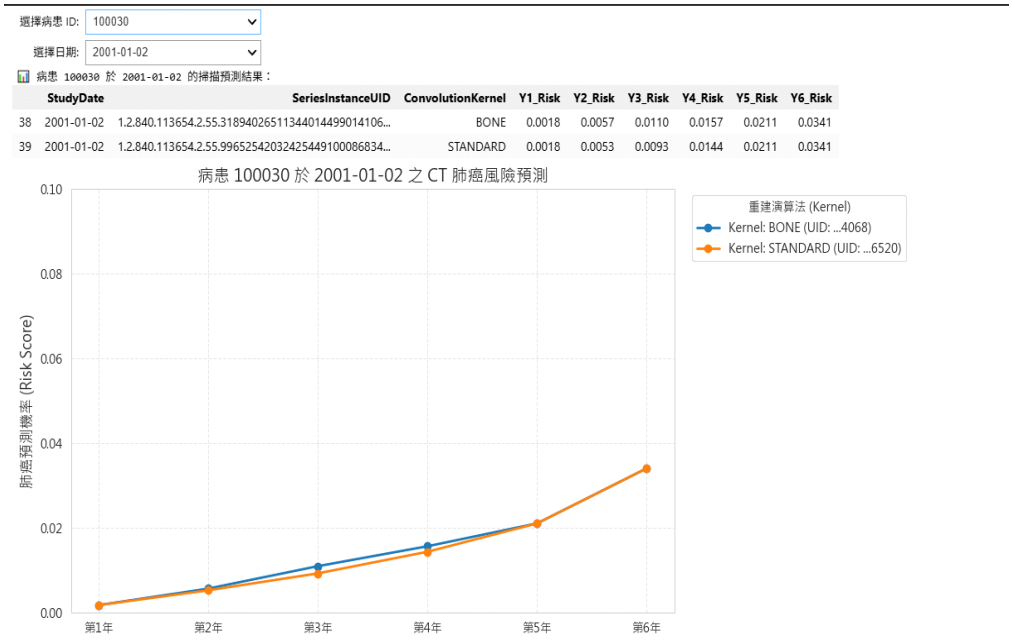
To standardize inputs for the Sybil AI model, rigorous filtering and prioritization criteria were applied to the CT dataset. The base dataset was restricted to original, axial CT series with a slice thickness of less than 5mm, explicitly excluding scout and localizer images. Because overly sharp reconstruction kernels can increase noise and artificially impact AI model predictions, a custom prioritization algorithm was applied for patients with multiple available series on a given study date. Series were ranked by kernel suitability: "Standard" kernels were prioritized first, followed by "Lung", "B30F", and "FC01". In cases where multiple series shared the highest available kernel priority, the sequence containing the highest total number of slices was selected. This automated hierarchy ensured the AI received the highest quality, lowest-noise volume available for every temporal timepoint.

Environment(三):

Operating System: Windows 10, CPU: AMD Ryzen 5 5600XT 6-Core Processor, RAM: 12GB RAM, GPU: NVidia 4060 8gb, python version: 3.9.25, Pyporch verion:2.7.1+cu118, Cuda version: 11.8, inference device: GPU, git version: git version 2.53.0. windows.2

Sybil Inference(三四):

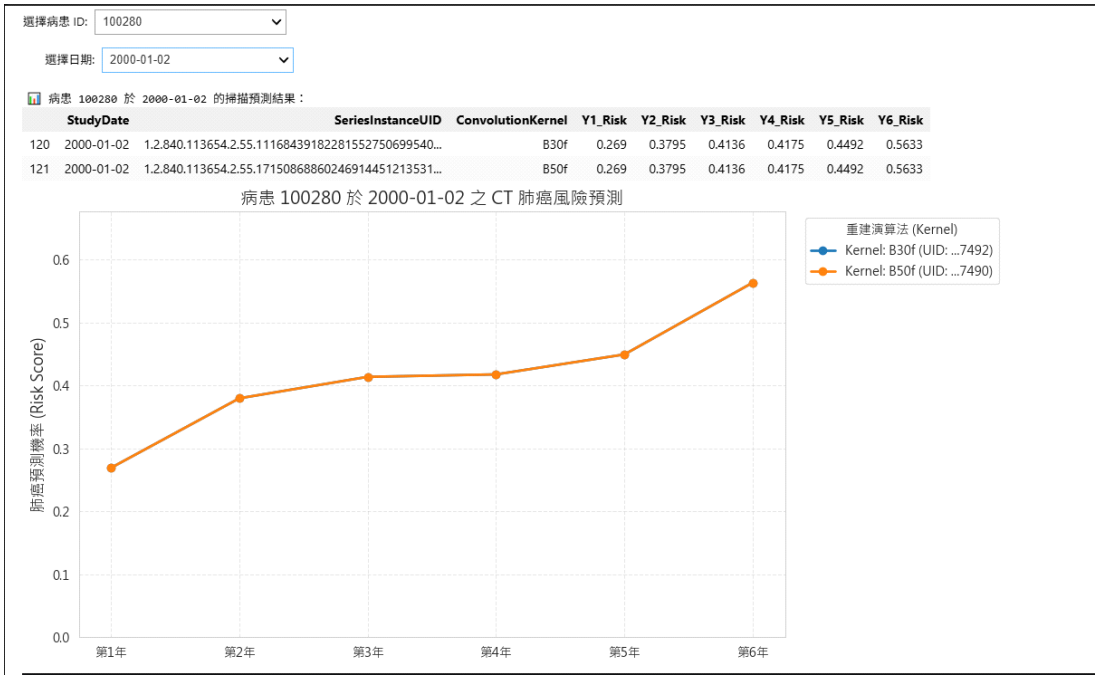
Results:


選擇病患 ID: 100030

選擇日期: 2001-01-02

病患 100030 於 2001-01-02 的掃描預測結果：

	StudyDate	SeriesInstanceUID	ConvolutionKernel	Y1_Risk	Y2_Risk	Y3_Risk	Y4_Risk	Y5_Risk	Y6_Risk
38	2001-01-02	1.2.840.113654.2.55.318940265113440144990141 06...	BONE	0.0018	0.0057	0.0110	0.0157	0.0211	0.0341
39	2001-01-02	1.2.840.113654.2.55.996525420324254491000868 34...	STANDARD	0.0018	0.0053	0.0093	0.0144	0.0211	0.0341


選擇病患 ID: 100280

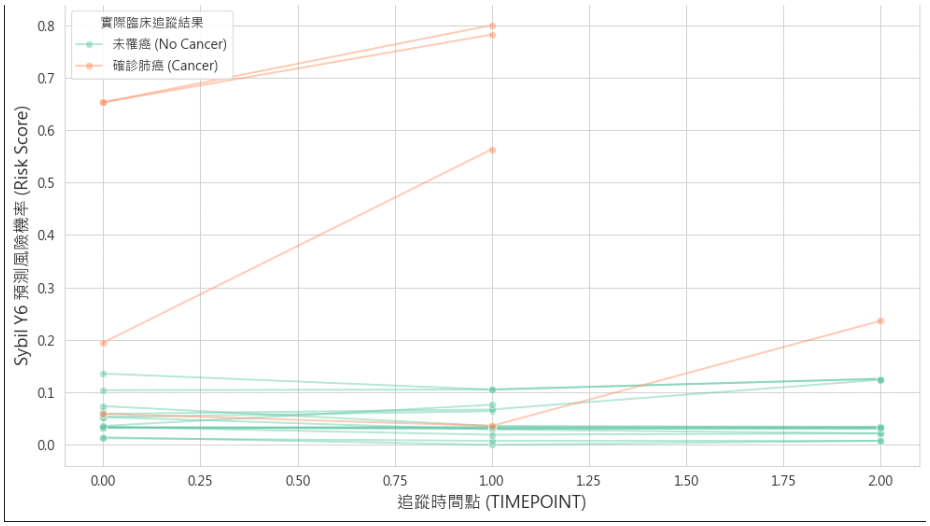
選擇日期: 2000-01-02

病患 100280 於 2000-01-02 的掃描預測結果：

	StudyDate	SeriesInstanceUID	ConvolutionKernel	Y1_Risk	Y2_Risk	Y3_Risk	Y4_Risk	Y5_Risk	Y6_Risk
120	2000-01-02	1.2.840.113654.2.55.111684391822815527506995 40...	B30f	0.269	0.3795	0.4136	0.4175	0.4492	0.5633
121	2000-01-02	1.2.840.113654.2.55.171508688602469144512135 31...	B50f	0.269	0.3795	0.4136	0.4175	0.4492	0.5633

The figures above illustrate the Sybil model inference results for 2 of the 19 evaluated patients. The line graphs map the predicted lung cancer risk trajectories, with the x-axis denoting the prediction timeframe (Years 1 – 6) and the y-axis representing the calculated risk score. Additionally, the accompanying data table in each figure details the CT study date, Series Instance UID, convolution kernel, and the exact Y1 to Y6 predicted risk values. (n = 19 patients, comprising of 122 series of CT scans total)

探索性分析:



The above figure presents the exploratory data analysis of CT scan predictions processed by Sybil. The x-axis denotes the sequential tracking timepoints for each patient, where 0.0 represents the baseline (index) CT scan, and 1.0 or 2.0 represent subsequent annual follow-up scans. The y-axis indicates the Sybil model's predicted probability (Risk Score) of the patient developing lung cancer within 6 years. Individual patient trajectories are color-coded by their actual clinical outcomes: orange lines represent patients who were eventually diagnosed with lung cancer, while green lines represent patients who remained cancer-free throughout the follow-up period (n = 19 patients. Note: For timepoints containing multiple CT scan series, such as different reconstruction kernels, the predicted risk scores were averaged to generate a single consolidated trajectory point).

	Y1_Risk	Y3_Risk	Y6_Risk
Mean (平均值)	0.030607	0.056039	0.094807
SD (標準差)	0.098471	0.139438	0.166503
Median (中位數)	0.001800	0.011000	0.034100
IQR (四分位距)	0.005700	0.021950	0.044100
Minimum (最小值)	0.000000	0.000000	0.000000
Maximum (最大值)	0.499400	0.699300	0.806000

The table above displays the mean, standard deviation (SD), median, interquartile range (IQR), minimum, and maximum risk probabilities generated by the Sybil model across the evaluated cohort. The increasing trend in both mean and maximum values over time reflects an accumulating risk for future lung cancer

development.

Batch Processing(五): 包含自動跳過以及暫存檔以利重跑。

External validation results example:

假設外部驗證得到下列結果。這些數值為情境題資料，請依已提供資訊進行判讀。

項目	假設結果
Sybil	AUROC = 0.72
Lung-RADS	AUROC = 0.74
Sybil＋clinical factors	AUROC = 0.77
Sybil calibration slope	0.71
Heavy smoker subgroup	AUROC = 0.79
Never／light smoker subgroup	AUROC = 0.61

Sybil initially scores lower than Lung-RADS; however, its predictive accuracy surpasses Lung-RADS when combined with clinical factors. This indicates that incorporating supplementary clinical information can significantly enhance the model's performance. Interestingly, Sybil demonstrates its highest accuracy within the heavy smoker subgroup. This suggests that the model is highly effective for patients with a history of smoking, though this is likely due to its training data predominantly featuring heavy smokers as positive cases.

Interestingly, the light smoker subgroup yielded the lowest scores, indicating that Sybil is less effective at predicting lung cancer risk for non-smokers and light smokers. It is important to note that the population demographics and environmental factors of the external validation cohort are unknown. This highlights a critical consideration, particularly in regions where environmental pollution is a significant concern, such as Taiwan (Hill et al., 2023). Furthermore, since genetic mutations are a primary driver of lung cancer, incorporating additional clinical data—such as Next-Generation Sequencing (NGS) for EGFR and ALK mutations—could serve as an effective method to increase predictive accuracy among light smoker subgroups. Ultimately, because Sybil was trained exclusively on the NLST dataset, which primarily consists of patients with heavy smoking histories, the model inherently faces limitations when applied to light-smoking populations.

Research Understanding (六、七)

1. Sybil 預測的是什麼 (What Sybil Predicts): Sybil predicts the risk of a patient developing lung cancer by utilizing AI-driven predictive models to analyze lung CT scans.

2. Sybil 與 Lung Cancer Detection, Lung Nodule Detection, Lung-RADS 及 Clinical Diagnosis 的用途有何不同?

- Sybil: Functions as an AI-based prognostic risk assessment tool to predict future lung cancer development.
- Lung Nodule Detection: Focuses strictly on identifying and pinpointing the spatial localization of specific lesions within the lung.
- Lung-RADS: Serves as a standardized classification system to interpret low-dose CT (LDCT) screening findings and guide follow-up management decisions.
- Clinical Diagnosis: Involves definitively confirming the pathology of lung cancer, which requires physical tissue extraction (biopsy) and examination.

3. 若某病例 Y6 risk 明顯高於其他病例, 可以得到哪些結論? 哪些結論目前不能成立?

A patient with a significantly higher predicted Y6 risk has a greater statistical probability of developing lung cancer within a six-year timeframe. However, it cannot be concluded that the patient currently has lung cancer; the score is strictly a prognostic prediction, not a definitive clinical diagnosis.

4. 若將模型由 NLST 應用至另一家醫院, 您認為最重要的潛在問題為何? 請說明原因。

The primary potential issue is the overestimation of lung cancer risk in healthy patients. Because the NLST training dataset predominantly consists of individuals with a heavy smoking history, the model may carry an inherent bias when applied to general or non-smoking populations. Additionally, Sybil struggles with the accurate spatial localization of attention maps. Furthermore, the model was trained on older NLST CT data; newer CT scanners produce sharper images with different texture patterns, which may introduce domain shift and decrease Sybil's predictive accuracy.

自主閱讀 (Literature Review)

Paper 1: Sybil: A Validated Deep Learning Model to Predict Future Lung Cancer Risk from a Single Low-Dose Chest Computed Tomography

- Research Question: To develop an AI model utilizing CT scans to predict the probability of lung cancer occurrence over a 1- to 6-year period.

- Study Population:
    - Training: NLST trial cohort (Caucasian individuals aged 55 – 74 with a heavy smoking history).
    - Validation: MGH and CGMH datasets (comprising both smokers and non-smokers).
- Model Input: A single baseline LDCT DICOM image.
- Outcome: The predicted probability of developing biopsy-confirmed lung cancer within 1 to 6 years following the index CT scan.
- Study Design: A retrospective cohort study utilizing 3D Convolutional Neural Networks (CNN) to identify sub-visual phenotypes from training data.
- Validation Strategy: External validation utilizing the MGH and CGMH datasets.
- Main Findings:
    - Achieved a 1-year lung cancer risk AUROC of 0.92 (NLST) and 0.94 (CGMH).
    - Maintained a 6-year cumulative risk C-index of 0.75 – 0.81 across all three validation sites.
    - Confirmed that LDCT images contain sub-visual, long-term prognostic signals that extend beyond human visual recognition capabilities.

Paper 2: External Testing of a Deep Learning Model for Lung Cancer Risk from Low-Dose Chest CT

- Research Question: To externally validate Sybil's ability to identify individuals at high risk for lung cancer within an Asian health checkup cohort.
- Study Population:
    - Overall: 18,057 individuals aged 50 – 80 undergoing opportunistic screening at a single medical checkup facility in Seoul, Korea.
    - Subgroups: Stratified into a heavy-smoking subgroup (≥20 pack-years, n=2,848) and a never/light-smoking subgroup (median 2 pack-years, n=9,943).
- Model Input: A single baseline LDCT scan, requiring no additional clinical information.
- Outcome: Incidence of lung cancer within 6 years following the baseline LDCT scan.

- Study Design: Retrospective cohort study.
- Validation Strategy: Dedicated external testing of the Sybil model without any fine-tuning or transfer learning. Predictive performance was evaluated using time-dependent AUC for 1 to 6 years, stratified by smoking history and baseline tumor visibility, with a matched subset comparison against Lung-RADS v2022.
- Main Findings:
    - Overall sample: Achieved a 1-year risk AUC of 0.91 and a 6-year risk AUC of 0.74.
    - Heavy-smoking subgroup: Demonstrated excellent performance for visible lung cancers (1-year AUC 0.94) and acceptable performance for future lung cancers (6-year AUC 0.70).
    - Never/light-smoking subgroup: Showed comparable performance for visible lung cancers (AUC 0.89) but poor discriminative performance for future lung cancers (AUC 0.56).
    - Localization: Attention maps demonstrated a limited ability to localize eventual tumor sites for future lung cancers not yet visible on the baseline imaging.

Paper 3: Role of Sex in Lung Cancer Risk Prediction Based on Single Low-Dose Chest Computed Tomography

- Research Question: To determine whether Sybil predicts lung cancer risk with equal accuracy regardless of patient sex.
- Study Population: 10,573 LDCT scans from 6,127 consecutive male and female lung cancer screening participants across a health system between 2015 and 2021.
- Model Input: A single baseline LDCT scan, requiring no additional clinical information.
- Outcome: Incidence of lung cancer within 6 years following the baseline LDCT scan.
- Study Design: Retrospective evaluation comparing predictive performance between female and male patients.
- Validation Strategy: Evaluation of Y1 to Y6 ROC curves, utilizing DeLong's test to compare male and female AUCs, alongside C-index calculations for prognostic valuation.
- Main Findings:

- Sybil predicted lung cancer risk in female patients with a high accuracy comparable to male patients.
- Notably, for the 6-year prediction timeframe, the AUC for females was higher than the AUC for males.

Limitations & Research Gap

- Poor Spatial Localization of Potential Lesions: Sybil's attention maps exhibit limited capabilities in localizing future tumors. A potential solution is to train the attention mapping utilizing longitudinal data (e.g., 3-year sequential CT scan records from patients diagnosed with cancer) to enhance spatial predictive accuracy.
- Low Accuracy in Non-Smoker Cohorts: Further investigation is urgently needed to determine why the model underperforms in predicting outcomes for non-smokers. Potential driving factors that need to be accounted for include unique genetic mutations and specific environmental exposures (such as air pollution).
- Unpredictability of Genetic Mutations: The model faces inherent limitations in predicting lung cancer onset driven by sudden or aggressive genetic mutations, which may not present early sub-visual phenotypic indicators on a baseline CT scan.

discussion

Incorporating Next-Generation Sequencing (NGS) to screen for EGFR mutations could potentially increase predictive accuracy; however, the high cost remains a significant barrier. In Asian populations, the EGFR mutation rate is notably high, ranging from 40% to 55% (Shi et al., 2014). Although Taiwan’ s National Health Insurance (NHI) recently began reimbursing NGS, this coverage is currently limited to patients already diagnosed with lung cancer, making it impractical for early screening. Alternatively, exploring epigenetic markers could be a cost-effective approach.

Blood-based biomarkers offer another non-invasive method to enhance prediction. While a major challenge is that biomarker concentrations in early-stage lung cancer are often too low to be detected by standard clinical assays, integrating them with imaging data shows promise. Recent studies have demonstrated that combining AI-driven CT models with biomarkers such as carcinoembryonic antigen (CEA), neuron-specific enolase (NSE), and cytokeratin 19 fragment 21-1 (CYFRA 21-1) significantly improves predictive accuracy compared to using AI alone **(Wang et al., 2025)**.

關鍵字:

LDCT: Low-Dose Computed Tomography gold standard for lung cancer screening. Serves as the sole input in sybil model.

AUROC: Statistics used to evaluate the performance of binary classification model, with value ranging from 0.5(random guess) and 1(perfect discrimination). Used to evaluate sybil model.

C-index: measurement to evaluate the predictive accuracy of a model regarding a time to event outcome. Used to evaluate the accuracy of prediction over the time.

Lung-RADS: standard classification system to assess lung nodules size and morphology, stratifying lung cancer risk and providing follow-up managing. Serves as a clinical base line for sybil to compare

Sub-visual phenotypes: micro textures or density variations etc. that are hidden within image, of which human visual recognition are useless while visible to computer or ai model. Sybil uses Therse subvisual phenotypes to train and predict lung cancer.

External validation: evaluating models using data not used in training the model. It serves to ensure sybil generalizability and applicability across diverse populations

Retrospective Cohort: study designed to use existing historic data to evaluate clinical outcome that have already occurred. Sybil uses these data to train or validate accuracy.

Overfitting: when ai models memorizes random noise or quirks of data instead of the actual underlying rule. If overfitted sybil can run perfect on trained data but performs bad with unseen data.

External Validation Proposal(八)

Data cohert:中國醫 lung LDCT scan obtained by hospital of age of 50-80years old or other hospital around Taiwan. Containing data of smokers and nonsmokers. Other information such as second-hand smoke exposure or kitchen usage or area of habitat or history of lung infections that might increase chances of lung cancer can easily be obtained using survey without other medical procedures. Addition younger patients LDCT scan might be useful since the age getting lung cancer are becoming younger in Taiwan (Wang et al., 2022) as well as east Asia (Chen et al., 2020).

Inclusion criteria: with at least 1 complete ldct scan with section under 5mm, and with e record.

Indext ct outcome and follow up: biopsy-confurmed incident lung cancer. Follow up tract 6y

Statistics analysis sybil model y1toy6 risk distribution, evaluating using auroc and c-index. Comparing with auroc and lung-RADS using Delongs test

Subgroup: smoking history or non smoking, kitchen or non kitchen , area of habitat, age, sex difference.

Restriction:hard for patients to follow up in the same hospital.

Negative or Unexpected Results:

If Sybil's performance on our hospital's dataset does not outperform the clinical baseline, it does not mean the study has failed. Under current regulations in Taiwan, Sybil can only serve as an assistive diagnostic tool for physicians, rather than a replacement. Furthermore, data shows an increasing incidence of lung cancer among young adults in Taiwan, whereas the US does not show a similar significant increase. This highlights that regional differences may affect Sybil's predictive capability. Additionally, factors such as CT image acquisition, slice thickness, and reconstruction algorithms can all influence the AI model's recognition. It is also possible that the sample size of lung cancer patients in Taiwan is relatively small compared to the US.

Subsequent subgroup analyses are required to determine whether the model failed completely, or if the issue lies primarily in predicting non-smokers, where its performance is already known to be weaker. Finally, we should investigate whether incorporating additional clinical information could improve the model's predictive accuracy.

.**10. Publication Plan**

1. Title When Imaging Phenotype and AI-Predicted Risk Disagree: Characterizing Discordance Between Sybil and Pulmonary Nodule Features

2. Target Journal European Radiology; Lung Cancer

3. Research Question Among patients undergoing low-dose CT (LDCT) screening, does the discordance between Sybil-predicted lung cancer risk and baseline pulmonary nodule characteristics provide additional prognostic information for future lung cancer development? Specifically, do patients with high Sybil-predicted risk but pure ground-glass nodules (pGGOs) have different subsequent lung cancer risks from patients with low Sybil-predicted risk but more suspicious nodule features, such as part-solid nodules or solid nodules?

4. Novelty Sybil estimates future lung cancer risk from a single LDCT scan, whereas pulmonary nodule characteristics provide more localized information about individual lesions. These two sources of information may not always agree. Rather than evaluating Sybil performance alone or stratifying patients solely by nodule type, this study will investigate whether discordance between AI-predicted risk and radiological nodule phenotype identifies clinically meaningful risk subgroups. The study will also assess whether combining Sybil scores with nodule characteristics improves risk stratification beyond either information source alone.

5. Introduction (Paragraph Planning)

- **P1:** Highlight the limitations of current nodule-based screening guidelines (e.g., Lung-RADS)(Pinsky, 2015) in risk stratification, especially regarding the indolent progression of pGGOs common in Asian populations.(Suzuki et al., 2011)
- **P2:** Introduce the clinical dilemma: Sybil can predict future risk, but what should clinicians do when "AI-predicted high risk" conflicts with a "low-risk baseline imaging phenotype"?
- **P3:** State the study objective: To determine which metric better reflects the true longitudinal risk when discordance occurs, and to evaluate the incremental value of combining both.

6. Methods

- **Study population:** Identify patients undergoing baseline LDCT screening with available Sybil predictions, pulmonary nodule characteristics, and longitudinal lung cancer follow-up data.
- **Imaging characterization:** Classify baseline nodules by type (pure ground-glass, part-solid, or solid), size, and solid-component characteristics.
- **Risk stratification:** Categorize patients according to Sybil-predicted risk and nodule phenotype. Compare concordant and discordant subgroups.
- **Statistical analysis:** Compare subsequent lung cancer incidence and time to diagnosis across subgroups using Kaplan–Meier curves and Cox proportional hazards models.
- **Sensitivity analyses:** Assess whether findings remain consistent after accounting for age, sex, smoking history, and nodule size.

7. Results (Figures and Tables Planning)

- **Table 1:** Baseline Demographics and Clinical Characteristics of concordant and discordant subgroups.

- **Table 2:** Predictive Performance (AUROC, C-index, and Calibration) comparison between single-source information and the combined model.
- **Figure 1:** Patient Selection Flowchart, including DICOM QC and phenotype data linkage.
- **Figure 2:** Kaplan–Meier Survival Curves comparing cancer-free survival across discordant subgroups (e.g., High Sybil + pGGO vs. Low Sybil + Solid).
- **Figure 3:** Representative Clinical Cases of discordance, showcasing Sybil risk trajectories alongside nodule phenotype evolution.

8. Discussion (Paragraph Planning)

- **Paragraph 1:** Summarize core findings: Which metric (AI or phenotype) serves as a better predictor when their risk estimations diverge?
- **Paragraph 2:** Explore the clinical and biological implications of the "High Sybil risk + pGGO" subgroup.
- **Paragraph 3:** Compare results with existing literature on Lung-RADS validation and discuss the reduction of false positives.
- **Paragraph 4:** Address clinical implementation: How discordance analysis can help personalize LDCT screening intervals.

9. Limitations

- **Nodule annotation:** Reliable classification requires radiological reports; DICOM metadata alone are insufficient.
- **Longitudinal follow-up:** Incomplete follow-up may lead to outcome misclassification.
- **Risk threshold selection:** Arbitrary Sybil cutoffs may affect subgroup definitions and require independent validation.
- **Generalizability:** Findings from a single cohort may not generalize to other populations with different screening practices.
- **Clinical interpretation:** Discordance does not necessarily mean either source is incorrect, as they describe different aspects of lung cancer risk.

10. Conclusion This study will determine whether discordance between Sybil-predicted risk and baseline pulmonary nodule phenotype identifies patient subgroups with distinct future lung cancer risks. If the combined model demonstrates improved clinical utility, it may support more individualized risk stratification.

Reference

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